1_Power

BUCK 12V – 4V5

2A time-delay JFC1032-1200TS

Fused 12 V input; legacy net name retained for existing copper.

BUCK 12V – 6V

BUCK 12V – 3V3

Input Protection

INPUT PROTECTION — REGULATED 12 V ONLY
±3.8 V clamp; ~2 A current limit; ~18 ms soft start
Validate startup, hot plug and simultaneous loads.

TPS259472ARPWR

3_PERIPHERAL

BUTTON GPIO TO ESP32

J19 pin 1: GND; pins 2–5: SW1, SW2, SW3, SW4.
Active-low; connect J9 Qwiic cable for HOST_3V3 pull-ups.
Suggested ESP32 GPIOs: 32, 33, 25, 26 respectively.

7-bit I²C address: 1 A5 A4 A3 A2 A1 A0 (R/W sent separately)

Open = 0 / Closed = 1

Valve channels 5/6: full ON / full OFF only.

J3: LOCAL DEBUG BUS (GND / +3V3 / SDA / SCL); onboard supply.
J9: dedicated standard Qwiic master port, J1 / J12 / J17: powered sensors.
Multiple boards: external TCA9548A at 0x70; one channel active.
Disable PCA9685 ALLCALL after initialization.

2_Pump / Valve Drivers

Rated voltage	DC 6.0V
Use voltage	DC1.8V~3.0V
Load current	<= 500mA

额定电压 (Rated voltage)	DC 6.0V
使用电压 (Operating voltage)	DC 1.8V~3.0V

Qwiic Master Interface

J9 TO USB / BATTERY ESP32

LOCAL A: RheoBoard +3V3 and sensor bus
HOST B: ESP32-supplied +3V3; regulators never paralleled
100 kHz initial bring-up; cable/module pull-ups share a current budget
J1 / J12 / J17 remain powered Qwiic sensor ports
J3 = LOCAL DEBUG BUS; no master-power isolation at that header

4_Switch Inputs

5_Pressure Sensor

H1 MountingHole · H2 MountingHole · H3 MountingHole · H4 MountingHole

Sheet: /
File: RheoBoard_v1.05.kicad_sch
Title:
Size: User | Date: | Rev:
KiCad E.D.A. 10.0.0 | Id: 1/1